

PHOENIX DEER VALLEY, AZ, USA

WMO: 722784

Lat: 33.688N

Lon: 112.082W

Elev: 444

StdP: 96.11

Time Zone: -7.00 (W07)

Period: 98-19

WBAN: 03184

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.6	4.1	-18.3	0.8	15.6	-16.0	1.0	16.5	8.7	12.6	7.6	13.2	1.4	20	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	42.4	20.5	41.3	20.3	40.1	20.0	23.7	33.8	23.2	33.4	22.8	33.4	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	17.2	27.2	20.6	16.2	27.7	19.1	14.7	29.2	73.1	33.8	71.2	33.1	69.4	33.1	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.4	7.3	6.1	-0.4	44.8	2.0	1.3	-1.8	45.7	-3.0	46.4	-4.1	47.1	-5.6	48.0	
			DB												
			WB	-3.6	24.7	1.6	1.6	-4.8	25.8	-5.7	26.7	-6.6	27.6	-7.8	28.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	12.9	14.0	17.6	21.1	25.8	31.4	33.8	32.9	30.2	23.8	17.3	12.1
	DBStd	8.39	3.34	3.43	3.74	3.73	3.75	2.83	2.38	2.39	2.87	3.64	4.00	3.21
	HDD10.0	36	12	6	1	0	0	0	0	0	0	0	1	16
	HDD18.3	634	170	124	59	17	2	0	0	0	5	65	193	
	CDD10.0	4705	102	119	236	332	490	643	738	711	606	427	221	82
	CDD18.3	2262	1	3	36	98	233	393	479	453	356	173	35	1
	CDH23.3	33875	18	59	443	1186	3230	6457	7914	7261	5149	1835	315	7
	CDH26.7	21069	0	3	118	473	1792	4337	5485	4857	3113	815	76	0
Wind	WSAvg	2.5	2.0	2.2	2.5	3.0	3.0	2.9	3.1	2.8	2.6	2.3	2.1	1.9
Precipitation	PrecAvg	200	22	20	18	9	3	3	19	33	17	16	13	25
	PrecMax	296	95	106	45	37	30	11	46	65	95	98	47	57
	PrecMin	103	0	0	0	1	0	2	2	8	0	1	0	4
	PrecStd	59	27	31	14	10	7	3	12	17	22	21	13	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.6	27.2	32.4	35.1	40.3	43.9	43.9	42.9	40.9	36.6	31.3	24.3
		MCWB	10.9	11.9	14.4	14.9	18.7	19.1	21.3	21.4	20.4	17.9	14.6	11.4
	2%	DB	23.1	25.3	29.5	33.0	37.8	42.0	42.5	41.6	39.1	34.3	28.8	22.2
		MCWB	10.4	11.3	13.1	14.4	17.4	18.7	21.0	21.1	19.9	17.1	13.5	10.8
	5%	DB	21.4	23.3	27.7	31.4	36.3	40.9	41.3	40.4	38.0	32.8	27.0	20.4
		MCWB	9.8	10.5	12.8	13.9	16.3	18.7	20.9	21.0	19.6	16.4	13.0	10.1
	10%	DB	19.5	21.3	26.0	29.5	34.5	39.2	40.0	39.0	37.1	31.3	24.9	18.5
		MCWB	9.1	10.2	11.9	13.0	15.5	18.0	20.7	20.9	19.5	15.9	12.2	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	14.3	15.3	17.9	20.6	22.8	24.6	25.5	23.5	21.4	16.7	14.3
		MCDB	18.3	20.2	28.1	30.9	37.9	37.6	35.4	36.5	32.6	26.8	23.3	18.3
	2%	WB	12.6	13.3	14.4	15.6	18.9	21.7	23.8	23.8	22.7	19.5	15.1	12.8
		MCDB	17.9	19.4	26.6	30.3	34.4	36.1	33.9	33.7	31.6	30.3	24.2	18.7
	5%	WB	11.5	12.3	13.6	14.5	17.7	20.8	23.3	23.3	22.0	18.3	14.0	11.6
		MCDB	17.6	19.2	24.9	28.7	32.9	35.7	33.6	33.5	32.0	29.2	24.3	17.5
	10%	WB	10.5	11.3	12.7	13.7	16.4	19.8	22.9	22.8	21.3	17.2	13.0	10.4
		MCDB	17.3	19.1	23.4	27.6	32.6	36.3	33.6	33.4	32.1	28.1	23.1	16.7
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.6	12.9	13.4	13.8	14.4	11.5	11.1	11.6	12.0	11.8	10.9
		MCDBR	13.8	14.7	15.4	15.5	15.7	15.6	12.8	12.3	12.6	13.4	13.8	13.0
		MCWBR	6.6	6.8	6.0	6.5	6.2	5.7	4.2	3.7	4.5	5.6	6.3	6.4
	5% WB	MCDBR	10.1	11.5	14.1	14.1	14.5	12.6	10.8	10.4	10.1	10.9	11.6	10.3
		MCWBR	5.3	5.5	5.6	6.3	6.3	5.4	3.2	3.2	3.9	5.1	5.8	5.4
Clear-Sky Solar Irradiance	taub	0.279	0.284	0.310	0.321	0.331	0.339	0.400	0.387	0.343	0.318	0.287	0.279	
	taud	2.582	2.580	2.484	2.438	2.445	2.420	2.378	2.402	2.500	2.522	2.564	2.588	
	Ebn at Noon	949	978	969	965	953	939	881	889	920	923	930	930	
	Edh at Noon	78	86	103	113	114	117	121	116	101	91	78	73	
All-Sky Solar Radiation	RadAvg	3.35	4.30	5.80	7.23	8.11	8.45	7.33	6.75	6.08	4.89	3.73	3.03	
	RadStd	0.26	0.34	0.30	0.22	0.30	0.32	0.35	0.24	0.23	0.29	0.16	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-110	N/A	N/A
(47)	Regional (13 neighbors)	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	-82	+107	+40

Nomenclature: See separate page